Entity-Relationship and Database Schema Architecture

Data Schemas and Relational Mappings for SceneCases, BuildingRisks, Tasks, Teams, and Assignments

SceneCases

Collection: flood_cases / xBD-S12

PK	**sceneId / caseId**	String (Indexed)
	disaster_name	String
	disaster_type	Enum (Flood, Hurricane)
	location (lat, lon)	GeoJSON Poi
	eventDate	Timestamp
	affectedBuildings	Integer
	floodPercentage	Float (0 - 100%)
	rasterTilePaths (SAR)	Array<String>
	status	Enum (active, reviewed)
	createdAt / updatedAt	Timestamp

1 : N (Contains)

BuildingRisks

Collection: building_risks / geojson

PK	**building_id**	Integer (Scene-Scope)
FK	**scene_id**	String (Ref: SceneCases)
	centroid_x, centroid_y	Float (Pixels)
	latitude, longitude	Float (WGS84)
	area_pixels	Integer (Footprint)
	flood_exposure_Ei	Float (0.00 - 1.00)
	damage_severity_Si	Float (0.00 - 1.00)
	risk_score_Ri	Float (0.00 - 1.00)
	priority_category	Enum (Critical, High..)
	footprint_polygon	GeoJSON Polygon

1 : 0..1 (Triggers)

RescueTeams

Collection: rescue_teams

PK	**teamId**	String (TEAM-xxx)
	teamName	String
	teamType	Enum (BOAT, USAR, DRONE)
	currentLocation (lat, lon)	GeoJSON Point
	available	Boolean
	personnelCount	Integer
	capabilities	Array<String>
	status	Enum (AVAILABLE, BUSY)
	speedKmH	Float
FK	**assignedTaskId**	String (Nullable Ref)

1 : N (Dispatched)

1 : N (Scopes)

RescueTasks

Collection: rescue_tasks

PK	**taskId**	String (TASK-xxx)
FK	**sceneId**	String (Ref: SceneCases)
FK	**buildingId**	Integer (Ref: BuildingRisks)
	location (lat, lon, area)	GeoJSON Point
	taskType	Enum (Rescue, USAR..)
	urgency	Enum (Critical, High..)
	riskScore	Float (0.00 - 1.00)
	taskReason	Text (Auditable Rationale)
	requiredResources	Array<String>
	taskStatus	Enum (PENDING, ASSIGNED)

1 : 0..1 (Allocated Via)

TeamAssignments

Collection: team_assignments

PK	**assignmentId**	String (ASSIGN-xxx)
FK	**taskId**	String (Ref: RescueTasks)
FK	**teamId**	String (Ref: RescueTeams)
	taskType	String
	distanceKm	Float (Haversine)
	allocationScore	Float (S_alloc)
	compatibilityReason	Text (Matching Logic)
	assignmentStatus	Enum (ASSIGNED..)
	assignedAt	Timestamp

DATABASE SCHEMA SPECIFICATION: PK Primary Key · FK Foreign Key Reference · 1 : N One-to-Many Relation · 1 : 0..1 Conditional Task Trigger · MongoDB + GeoJSON Models